

VGPA Test Plan: What We Run the Day Their Code Drops

Prepared September 12, 2026 — VGPA = Khatib, Galun, Basri, ECCV 2026 (arXiv:2609.09491)
Code status: project page (vgpa-sfm.github.io) shows a placeholder link; polled every results cycle

Why these tests

VGPA is label-free view-graph pose averaging from the same lab: it beats RESfM on most scenes under its own lenses, but every one of its numbers is a single run, in the published-RESfM environment (its main-table RESfM column is the paper’s 0.169 draw — verified by recomputing their column means), on its own track preprocessing, and never under heavy contamination, never on Olsson, never at 60%. Each test below closes one of those gaps. Ordered by cost and by decision value; every analysis lands in the already-built five-lens machinery (`compute_vgpa_metrics.py` + `analyze_vgpa_style.py`, `FIVELENS_readout.pdf`).

Tier 1 — cheap, immediately publishable (days)

1. **Released checkpoint, our environment, repeated runs.** Their pipeline includes test-time fine-tuning and robust BA — both nondeterminism sources we have quantified for our own arms (TTT ~ 0.14 mean-level; BA floor $\mathcal{O}(0.05)$). Three-plus repeats of their released model on their benchmarks measures *their* run-to-run variance. Their appendix already concedes BA nondeterminism; this puts a number on it. *Decision value: extends the environment-noise finding to a second method; rebuttal ammunition regardless of outcome.*
2. **Five-lens readout of their released model on their tracks.** Per-scene wins, mean-vs-median with blow-up attribution (their MegaDepth mean is carried by four scenes: 0265, 1001, 0024, 0104), coverage, AUC/RA-AUC recomputed under one protocol. Costless once (1) runs.

Tier 2 — the decisive test (weeks)

3. **Five-seed retrain with their recipe.** Training is self-supervised on MegaDepth — no label infrastructure needed. Tests whether the two phenomena we document for RESfM generalize: (a) the *checkpoint-provenance gap* (released vs. recipe-reproducible); (b) the *training-seed lottery* (bands, bimodality, catastrophic scene cells). *Decision value: if either appears, the paper’s reliability methodology upgrades from “about RESfM” to “about deep SfM training” — the strongest possible answer to a “VGPA supersedes this” review. If neither appears, VGPA earns a genuine robustness credit and the composition case strengthens.*
4. **VGPA on OUR track suite** (same npz benchmarks as every row of our tables): MegaDepth, 1DSfM at 43%, 1DSfM-hard at 60%, Strecha, BlendedMVS, Olsson. Requires adapting their view-graph construction to consume our tracks or rebuilding their front-end on our scenes — scope known only after reading their code. Two questions their paper cannot answer: *does the view-graph modality escape the Olsson collapse* (and specifically the dtu106 basin — our tail decomposition

shows that one scene is the collapse), and *does their two-view front-end survive 60% contamination*, where pairwise geometry estimation itself degrades — the label-ceiling band where our label-free removal is the only working recipe.

Tier 3 — composition (after Tiers 1–2)

5. **VGPA-init for the collapse basin.** Use VGPA’s global poses to initialize or anchor the joint reconstruction on the catastrophic cells (dtu106; the Olsson collapse seeds; BMVS bimodal seeds). *Caution: Fadi reports VGPA+RESfM composition underperformed — get the configuration details from him first (which coupling, which datasets, worse than what). Our probe differs on both axes he presumably tested: targeted at basin-failure cells rather than benchmark-wide, and composed with the self-supervised arms rather than supervised RESfM.*
6. **End-to-end label-free pipeline.** VGPA poses + our track-level adaptive cleaning + robust BA: both stages self-supervised — the composition both papers’ future-work sections gesture at. Report under both metric families (raw error over registered + RA-AUC), since our lens-4 results show the two can invert verdicts.
7. **Retrieval-graph stress test.** Their MegaLoc@30 sparse-graph variant on contaminated scenes: sparse graphs amplify the cost of bad edges; our contamination axis is the natural stressor their Table 5 never meets.

Venue timing

Code before ~Oct 20: Tier 1 (+ Tier 2a if fast) can enter the CVPR submission. Code before ~mid-Jan: Tiers 1–2 are rebuttal material. Later: Tiers 2–3 seed paper #2 (with the VGPA composition conversation as the collaboration on-ramp with Fadi/Ronen).

Standing infrastructure (ready today)

Five-lens machinery validated on our full matrix; RA-AUC/AUC pipeline recovers poses from COLMAP models without re-evaluation; 1DSfM images downloaded (14 scenes); MegaDepth undistorted substrate built (68 scenes); the with-seeds protocol, verify scripts, and the paper’s environment-noise decomposition give every number above a place to land.